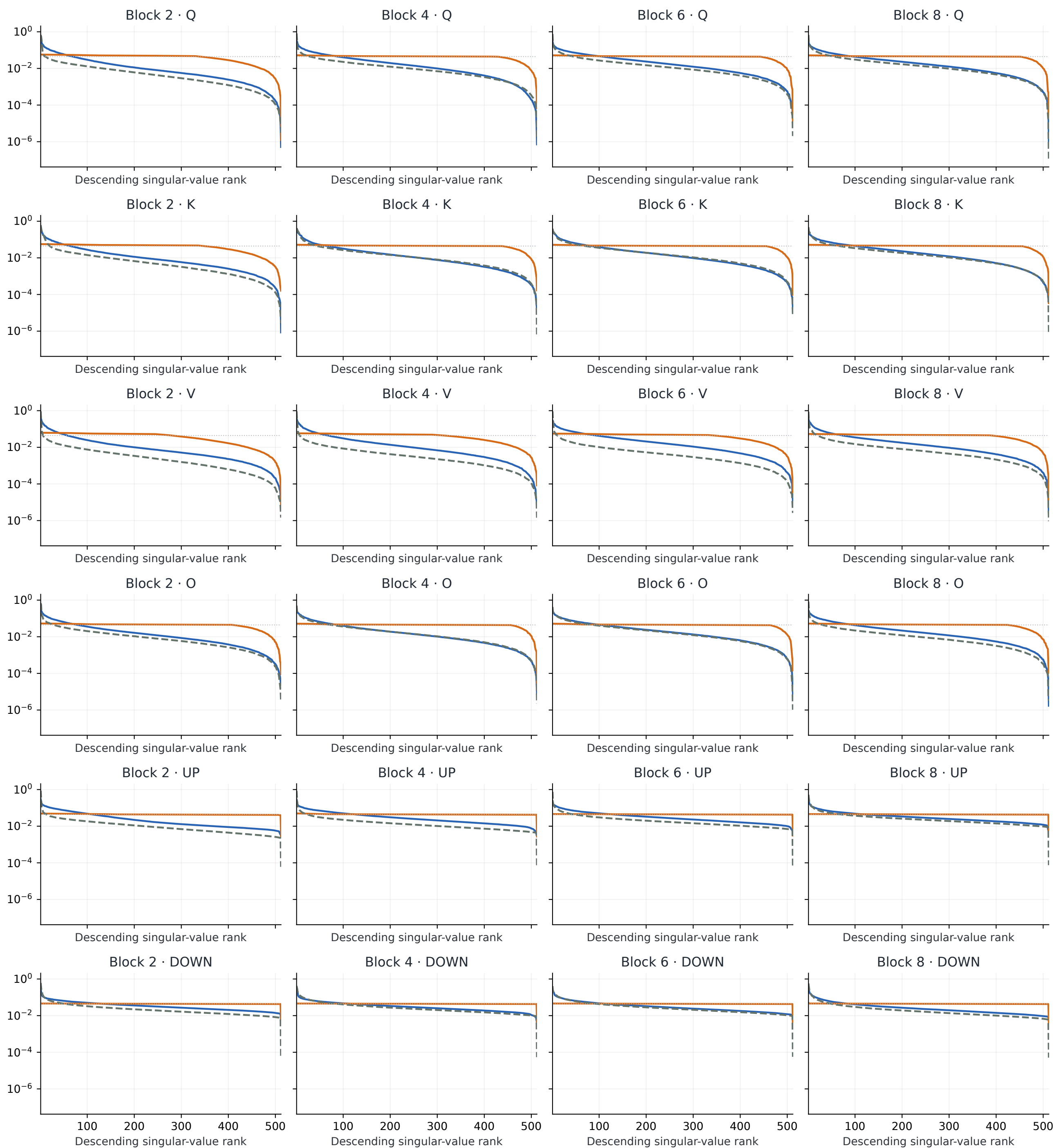


Full spectra at the final update · step 1469

AdamW adaptive update Muon post-NS update Muon pre-NS momentum



All 512 singular values are shown. Each matrix has unit Frobenius norm; leading singular values are retained.